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A
MID-TERM PROJECT REPORT
ON
**EDUMENTX: LOCATION BASED TUTOR DISCOVERY
PLATFORM**

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
ASCILITE	Australasian Society for Computers in Learning in Tertiary Education
BaaS	Backend-as-a-Service
CBS	Central Bureau of Statistics (Nepal)
DFD	Data Flow Diagram
GPS	Global Positioning System
GUI	Graphical User Interface
HTTP(S)	Hypertext Transfer Protocol (Secure)
IDE	Integrated Development Environment
IOE	Institute of Engineering
JSON	JavaScript Object Notation
KNN	K-Nearest Neighbors
LBS	Location-Based Service
LLM	Large Language Model
MFA	Multi-Factor Authentication
NCE	National College of Engineering
NLP	Natural Language Processing
NoSQL	Not only SQL
OTP	One-Time Password
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SEE	Secondary Education Examination (Grade 10, Nepal)
TU	Tribhuvan University
UI	User Interface
UX	User Experience

1. INTRODUCTION

1.1 Background

Education is one of the most important foundations for personal growth and social progress. Around the world, families invest heavily in their children's education, recognizing that formal schooling alone may not always be enough. When students struggle with certain subjects or need extra preparation for exams, many families turn to private tutoring as a way to fill the gaps. This practice of hiring tutors outside of regular school hours has become extremely common, as parents look for additional academic support to help their children perform better.

Private tutoring, commonly referred to as shadow education, has become a widespread practice as formal schooling systems often fail to meet the diverse learning needs of students [1]. This form of supplementary education has grown particularly fast in developing countries, where public school resources are limited and teacher-student ratios remain inadequate [2]. Research on shadow education as a global phenomenon confirms that it has expanded across all continents and income levels, becoming a parallel curriculum that operates alongside formal schooling [3].

Nepal's education system has made notable progress over the past decade. According to the National Population and Housing Census 2021, the overall literacy rate reached 76.3 percent among the population aged five years and above, up from 65.9 percent in 2011. The country has also experienced rapid urbanization during this period, with the urban population share jumping from 17.07 percent in 2011 to 66.17 percent in 2021 [4]. Despite these improvements, challenges within the mainstream education system persist. Overcrowded classrooms, inconsistent teaching quality, and limited resources in public schools continue to push families toward private tutoring. In cities like Kathmandu, dual-income households have become increasingly common, leaving working parents with limited time for direct academic supervision of their children. As a result, many urban families look for home tutors as a practical solution.

The phenomenon of shadow education is deeply rooted in Nepal's educational landscape. Thapa's study on private and public school competition in Nepal highlights how supplementary tutoring has become an established practice, with families across income levels

relying on it to fill gaps left by formal schooling [5]. Studies further indicate that approximately 68 percent of Grade 10 students in Nepal receive some form of private tutoring, confirming how widespread this practice has become at the secondary level.

Despite this growing demand, Nepal's tutoring market remains fragmented and informal. Parents and students typically rely on word-of-mouth recommendations or unstructured social media groups to find tutors. These methods offer no verification of tutor credentials, no proximity-based filtering, and no intelligent matching. With smartphone penetration in Nepal reaching 65 to 71 percent, the digital infrastructure exists to address these problems through a dedicated mobile platform.

This mid-term report presents EdumentX, a location-based mobile application designed to modernize home tutor discovery in Kathmandu Valley. The platform enables students and parents to find nearby verified tutors through an interactive map, enroll in long-term monthly plans, and receive personalized recommendations through an AI-powered RAG chatbot.

1.2 Problem Statement

The process of finding a home tutor in Nepal is currently slow, unreliable, and frustrating for everyone involved. Both students seeking academic help and tutors looking for teaching opportunities face significant difficulties due to the lack of any organized system.

The core problems are:

- (i) There is no way for parents to check whether a tutor is actually qualified or trustworthy before hiring them. Most families rely on recommendations from people they know, which limits their options and gives no real measure of teaching quality or safety.
- (ii) Independent tutors, including university students looking for part-time work, retired teachers, and subject specialists, have no proper platform to advertise their services. Many end up going through agencies that take large commissions, while others simply cannot find students outside their own neighborhood.
- (iii) The tools people currently use to find tutors, mainly WhatsApp groups and Facebook pages, are not designed for this purpose. Posts are unorganized, location details are vague (just area names with no actual distance), there is no way to filter or search, and there is no system to verify anyone's identity or qualifications.

At present, no mobile application exists in Nepal that lets users find tutors on a map, filter by distance, or get AI-based suggestions. EdumentX aims to solve these problems by offering proximity-based search on an interactive map, a verification system where admins review tutor documents and award Blue Tick badges, a structured monthly enrollment process, and a chatbot that pulls live tutor data to give personalized recommendations.

1.3 Objectives

- To develop a location-based mobile application that enables students and parents to discover and connect with nearby home tutors in Kathmandu Valley using map-based proximity search.
- To integrate an AI-powered RAG chatbot that allows users to find suitable tutors through natural language queries grounded in live platform data.
- To implement a tutor verification system that helps parents make safe and informed hiring decisions.

1.4 Scope

EdumentX is a role-based mobile platform for connecting students and parents with home tutors based on location, subject, and monthly enrollment preferences. The scope of this project includes the following:

- **Target Users:** Students, parents, independent home tutors, and platform administrators.
- **Geographic Focus:** Kathmandu Valley, where smartphone penetration and internet connectivity are sufficient to support the platform.
- **Service Type:** Physical (in-person) home tuition services only; no online or video-based classes.
- **Academic Levels:** School and college students at SEE, +2, and undergraduate levels.
- **Platform:** Cross-platform mobile application for Android and iOS built with React Native (Expo).
- **Discovery Methods:** Map-based search with proximity ranking and an AI chatbot for natural language queries.
- **Enrollment Model:** Long-term monthly enrollment plans (1, 3, 6, or 12 months) managed through structured request and response flows.

- **Verification:** Blue Tick badge system for tutor credential verification by admin.
- **Group Learning:** Support for group tuition batches created by tutors for shared, cost-effective learning.

2. LITERATURE REVIEW

2.1 Related Work

This section reviews recent academic research on location-based tutor matching systems, AI-powered educational platforms, and identifies the gaps that EdumentX aims to address.

A 2022 study presented at ASCILITE introduced MENTOR, an intelligent mobile peer tutoring application designed for both face-to-face and remote tutoring [6]. The system focused on matching peer tutors with students based on subject expertise and availability. While it demonstrated the value of mobile platforms for tutoring, it did not include any location-based features such as map visualization or proximity ranking. The matching was based solely on academic parameters without considering geographic distance between users.

A 2026 study published in the Journal of Emerging Technologies and Innovative Research described a Flutter-based mobile application for finding local tutors in India [7]. The system used Google Maps API for proximity-based tutor search and Firebase as the backend database. Students could filter tutors by subject, availability, and location. The application calculated distances between users using the Haversine formula and included features for booking and rating tutors. However, the researchers noted that AI-based recommendations were left as future scope. The entire matching process relied on manual filtering by the user, with no intelligent ranking or conversational search capability.

In 2025, researchers from Malaysia developed iTeachU, a system that combined AI with geolocation for tutor matching [8]. The platform used content-based filtering along with the K-Nearest Neighbors algorithm, incorporating Haversine distance calculations to rank tutors by both preference match and physical proximity. It was built using Flutter for the frontend and Flask for the backend. While the system demonstrated effective ranking within the Melaka region, its AI component was entirely rule-based. Users could not interact with the system using natural language. Instead, they had to select from predefined filters and dropdown menus, making the experience less flexible than a conversational interface.

All of these studies represent important steps toward technology-driven tutor discovery, but several gaps remain when considering the Nepali context. No existing research or

platform provides geospatial visualization of tutors on interactive maps with service area coverage for Nepal. City-level dropdown menus, which are used by most platforms, are not useful in dense urban areas like Kathmandu where a tutor two kilometers away is very different from one ten kilometers away. Independent part-time tutors such as university students and retired teachers lack a proper digital channel to reach nearby students, as existing systems tend to favor agency-managed tuition centers. None of the reviewed systems offer a natural language interface where users can simply type what they need instead of navigating through multiple filter options. While some platforms claim AI-based matching, none combine proximity, qualifications, availability, and budget into a single intelligent ranking powered by a large language model. The current methods available in Nepal, primarily social media groups, require significant manual effort to browse posts, contact agencies, and compare options without any automated assistance.

EdumentX brings map-based tutor visualization, a conversational interface for natural language queries, a structured verification system with Blue Tick badges, and a focus on empowering independent tutors to connect with nearby students efficiently.

2.2 Related Theory

2.2.1 Location-Based Services

Location-Based Services deliver context-aware services using a device’s real-time geographic position obtained through GPS, cellular triangulation, or Wi-Fi. In EdumentX, location-based services power the proximity-based tutor discovery and map visualization features. Tutor coordinates are stored as Firestore GeoPoint fields, which allow geospatial queries to be executed directly from the client application.

2.2.2 Haversine Distance Formula

The Haversine formula computes the great-circle distance between two points on Earth’s surface, accounting for the planet’s curvature. EdumentX uses this formula to rank tutors by their true distance from the student. The computation proceeds as follows:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (2.1)$$

$$c = 2 \arctan 2(\sqrt{a}, \sqrt{1-a}), \quad d = R \cdot c \quad (2.2)$$

where ϕ_1, ϕ_2 are latitudes in radians, λ_1, λ_2 are longitudes in radians and $\Delta\phi$ and $\Delta\lambda$ are

the coordinate differences, $R \approx 6,371$ km, and d is the output distance. Degree values from GeoPoints are converted using $\phi_{\text{rad}} = \phi_{\text{deg}} \times \frac{\pi}{180}$. For the 1 to 20 km range typical in Kathmandu Valley, this formula provides meter-level accuracy.

2.2.3 Geofencing and Radius-Based Filtering

Before applying the Haversine formula, EdumentX uses a rectangular bounding box pre-filter to narrow down the candidate set and reduce computation [9]. For a given search radius r and student position (ϕ_s, λ_s) :

$$\Delta\phi_{\text{max}} = \frac{r}{R} \cdot \frac{180}{\pi}, \quad \Delta\lambda_{\text{max}} = \frac{r}{R \cdot \cos(\phi_s)} \cdot \frac{180}{\pi} \quad (2.3)$$

Only tutors whose coordinates fall inside this bounding box proceed to the exact Haversine distance check, which keeps query response times low.

2.2.4 K-Nearest Neighbors Algorithm

KNN is a non-parametric algorithm used in EdumentX to rank tutors by overall suitability. Each tutor is represented as a feature vector $\mathbf{t} \in \mathbb{R}^5$:

$$\mathbf{t} = [d, r, q, e, \bar{v}] \quad (2.4)$$

where d is the Haversine distance, r is the monthly rate, q is the qualification score, e is years of experience, and $\bar{v}$ is the average rating. All features are scaled to the range $[0, 1]$ using min-max normalization: $x_{\text{norm}} = \frac{x - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}$. The similarity between a student query vector $\mathbf{s}$ and a tutor vector $\mathbf{t}$ is measured by weighted Euclidean distance:

$$D(\mathbf{s}, \mathbf{t}) = \sqrt{\sum_{j=1}^n w_j \cdot (s_j - t_j)^2}, \quad \text{Top-}k = \arg \min_{\mathbf{t} \in \mathcal{T}} D(\mathbf{s}, \mathbf{t}) \quad (2.5)$$

The default value of $k = 5$. Feature weights are assigned based on their importance for home tuition selection:

Table 2.1: KNN Feature Weights for Tutor Recommendation

Feature	Symbol	Weight w_j
Geographic Distance	d	0.35
Monthly Rate	r	0.25
Qualification Score	q	0.20
Average Rating	$\bar{v}$	0.15
Experience (years)	e	0.05

2.2.5 Retrieval-Augmented Generation

Retrieval-Augmented Generation is an approach that combines information retrieval with text generation to produce responses grounded in actual data rather than relying solely on a language model’s training data. The original RAG framework was introduced by Lewis et al., who showed that combining a pre-trained retriever with a sequence-to-sequence generator significantly improves accuracy on knowledge intensive tasks [10]. In EdumentX, RAG is used to retrieve matching tutor profiles from Firestore based on the user’s query and inject them as context into the language model’s prompt, ensuring that recommendations are based on real, live platform data. A recent survey on RAG chatbots in education confirms that this approach reduces hallucination and improves the relevance of generated responses in educational contexts [11].

2.2.6 React Native and Expo

The EdumentX mobile app is built with React Native, which lets a single codebase produce native iOS and Android binaries. Expo sits on top of React Native and provides a managed toolchain for device features such as location, camera, and push notifications, along with over-the-air updates that shorten the release cycle.

2.2.7 Firebase and Supabase Storage

Four services form the EdumentX backend. Authentication handles email/password and Google sign-in. Firestore is the main NoSQL store; tutor locations are saved as GeoPoint fields, and Security Rules control who can read or write which documents. Supabase Storage (1 GB free, no API key needed) holds verification documents and demo videos. The Firebase Admin SDK is used out-of-band from a local Node.js script to grant the administrator role to a fixed list of emails, since the `admins` collection is locked to Admin-SDK writes only.

The project does not rely on Cloud Functions. Distance ranking, the KNN scoring, and the retrieval step of the chatbot all run on the client. This keeps the build within the free Spark tier and avoids the paid Blaze plan that the team cannot sign up for.

2.2.8 OpenStreetMap and Open Geocoders

To avoid the paid Google Maps Platform API, EdumentX uses OpenStreetMap raster tiles for map rendering and the keyless Nominatim service for forward geocoding (turning a typed city name into latitude and longitude). Distance and KNN matching are computed on the device, which is fast enough at the Kathmandu-Valley scale where the target database size is in the low thousands of tutors.

3. METHODOLOGY

3.1 System Design

3.1.1 System Architecture

Figure 3.1 shows the overall system architecture of EdumentX. The architecture has three layers: the client side, the serverless backend, and external services.

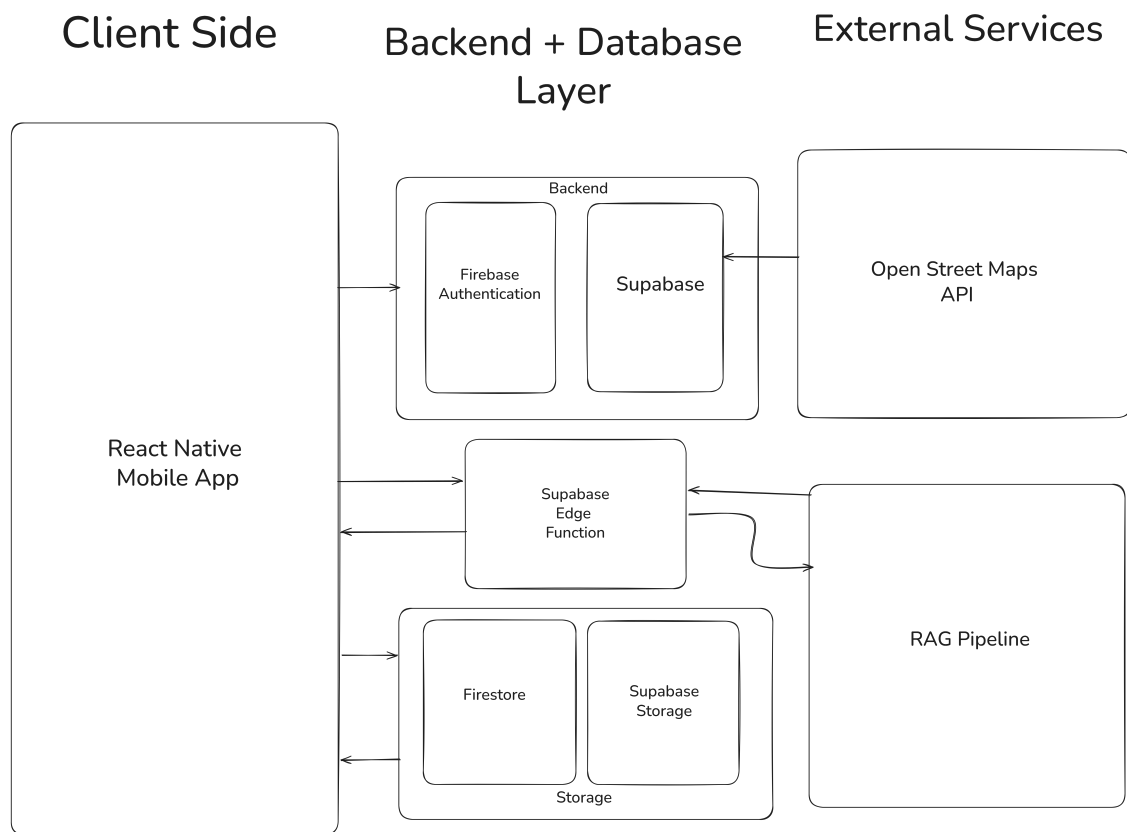


Figure 3.1: System Architecture of EdumentX

The client side is the React Native mobile application built with Expo. It serves as the single entry point for all users and communicates with the backend using the native Firebase SDK.

The serverless backend runs entirely on Firebase with three services. Firebase Authentication handles user registration and email/password plus Google sign-in. Cloud Firestore

stores all structured data including user profiles, tutor profiles, enrollment requests, and reviews, with tutor locations stored as GeoPoint fields and protected by Security Rules. Supabase Storage (1 GB free tier) holds tutor verification documents and demo videos.

The external services layer includes OpenStreetMap tiles for map rendering, Nominatim for keyless geocoding, and a free-tier LLM endpoint for the chatbot. EdumentX does not depend on Cloud Functions, Google Maps SDK, or any paid LLM; every external call is either keyless or fits within a free quota.

3.1.2 Use Case Diagram

Figure 3.2 shows the use case diagram with three actors: Students/Parents, Tutors, and Admin.

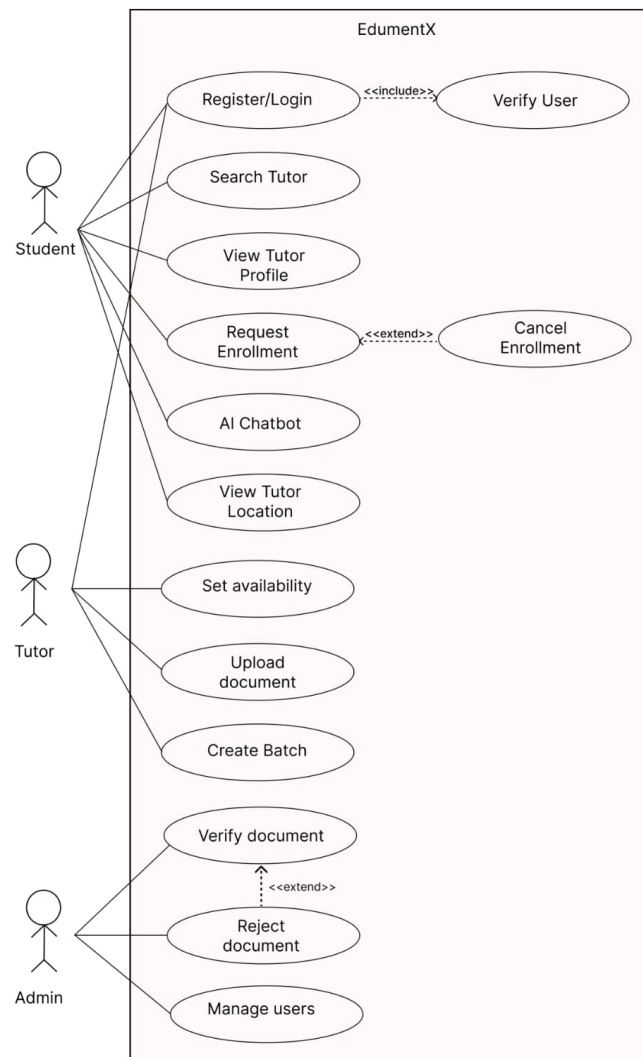


Figure 3.2: Use Case Diagram

Students can register, sign in, search for tutors using filters or the AI chatbot, view tutor profiles, send enrollment requests, cancel an active enrollment, and rate tutors after completing a plan.

Tutors can register, set their availability, upload documents for verification, manage their profile, respond to enrollment requests, and create group batches for shared tuition.

Admin can log in to the admin panel, verify or reject tutor documents, manage user accounts, and grant Blue Tick badges to verified tutors.

3.1.3 Data Flow Diagram

Figure 3.3 shows the Level 0 Data Flow Diagram. Users interact with the EdumentX system by sending requests and receiving responses. The system processes these requests and stores or retrieves data from Firebase Firestore.

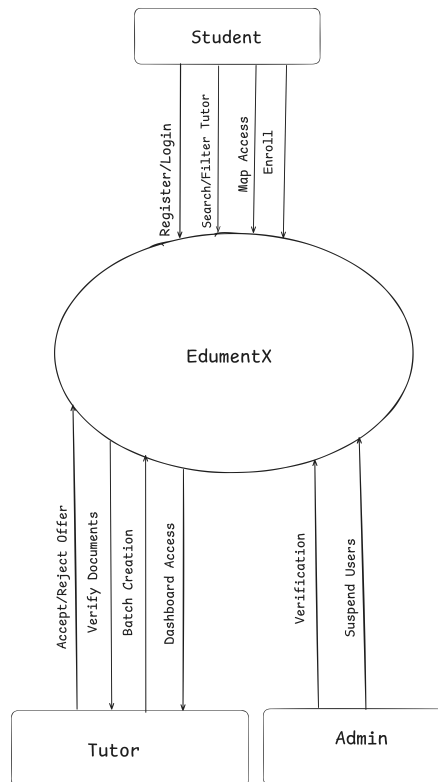


Figure 3.3: Level 0 Data Flow Diagram

Figure 3.4 shows the Level 1 Data Flow Diagram with detailed system processes including authentication, tutor search, AI chatbot recommendation, enrollment management, tutor verification, and rating handling.

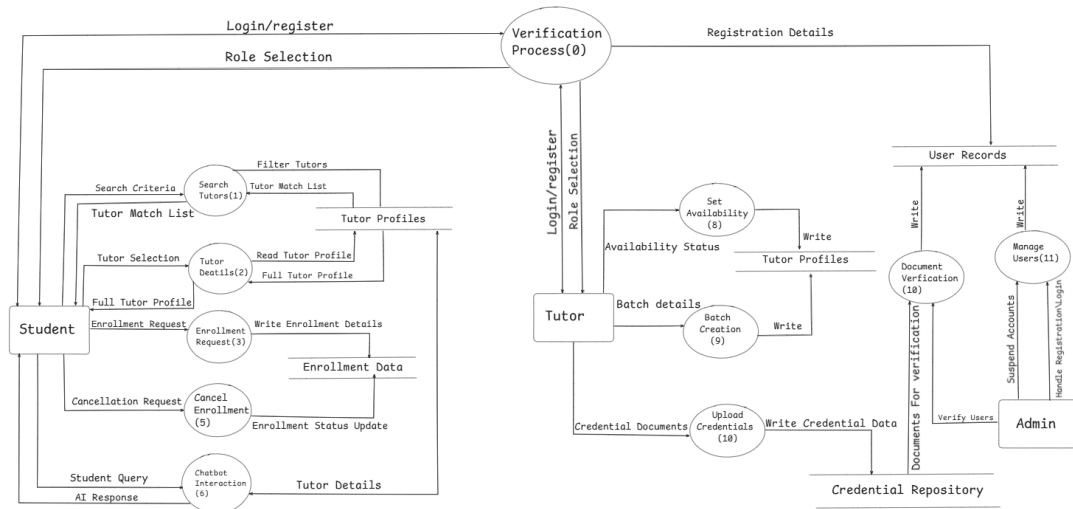


Figure 3.4: Level 1 Data Flow Diagram

3.1.4 RAG Chatbot Workflow

Figure 3.5 illustrates the RAG workflow used by the EdumentX AI chatbot.

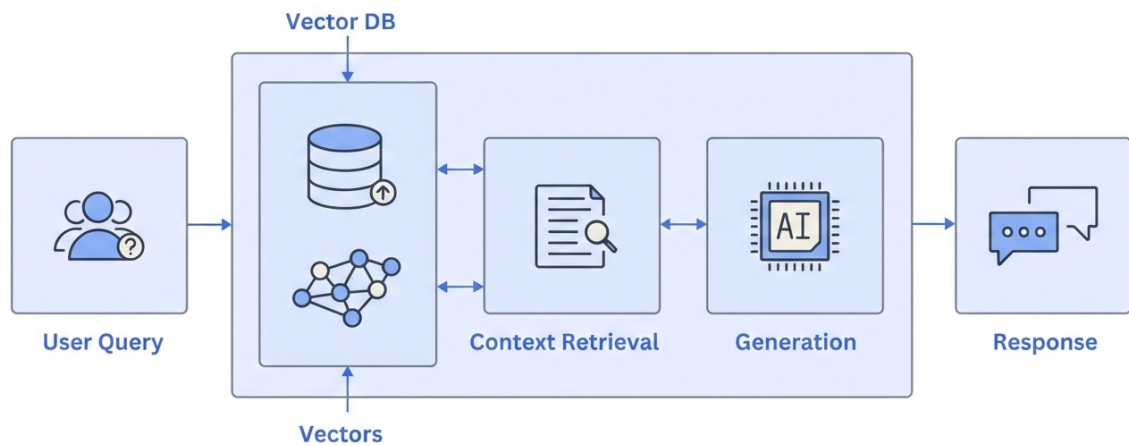


Figure 3.5: RAG-Based AI Chatbot Workflow

The chatbot works through a six-step pipeline. First, the student types a natural language query like “I need a science tutor near Lalitpur for 3 days a week under Rs 5,000.” The full conversation history is included so the chatbot can handle follow-up questions. Second, the client extracts structured parameters from the query: subject, location, budget, days per week, and plan duration. The free-tier LLM endpoint handles this extraction. Third,

the client queries the Firestore tutors collection, applies the bounding-box pre-filter followed by exact Haversine distance, and applies budget and subject filters. Fourth, the top ten matching tutor profiles are formatted and injected into the LLM's prompt as context data. Fifth, the language model generates a response recommending three to five tutors with name, rate, distance, rating, and verification status. Sixth, the response is rendered as a chat message in the app. Identical queries are cached for 15 minutes to keep the free-tier token budget healthy.

4. TASK ACCOMPLISHED

EdumentX is a mobile-first tutor discovery platform for Kathmandu Valley that helps students and parents find nearby verified home tutors through an interactive map, an AI-powered chatbot, and a structured monthly enrollment process. The following tasks have been accomplished till now as part of the EdumentX mid-term development.

4.1 Project Setup and Documentation

A solid planning base was set up before any code was written. A feature inventory and folder structure for documentation, reference material, and archived work were created so that the team could move in a single direction. A team change log and a project status document are maintained alongside the repository.

4.2 Design System

The application's visual design is built around a small set of design tokens for colors, surfaces, borders, and text that are registered in the build configuration. Shared UI primitives such as the button, card, screen, and header have been extracted for reuse. The styling layer was migrated off the previous Tamagui setup to plain NativeWind class names, which removed a large dependency surface and made the codebase easier to maintain.

4.3 User Sign-in and Sign-up

EdumentX supports two free sign-in methods. Email and password registration and login are wired through native Firebase Authentication, and Google sign-in is wired through the native Google Sign-In SDK. After an email sign-up, the user is shown a “check your inbox” panel that stays visible until the email is verified. An “I’ve verified” button on that panel refreshes the session so a user who returns from the inbox is not stuck on the pending screen. A routing guard in the root layout sends unverified users back to the sign-up screen, users without a role to the role selection screen, and verified users onward to their dashboard. A bug where an existing user was being routed back to role selection on

every sign-in was identified and fixed.

4.4 Onboarding and Profile

After sign-in, a new user lands on a role selection screen that writes the chosen role to the user record and routes the user to the matching onboarding form. The student onboarding form captures name, phone, city, grade, and subjects. The tutor onboarding form captures name, phone, city, headline, bio, monthly rate, subjects, and years of experience. A minimum length check on the city field prevents the form from being submitted with a one-character placeholder. Profile photos are uploaded to object storage and the resulting URL is persisted to the user record so that the photo survives an app restart.

4.5 Dashboards

The student home and tutor home screens are wired to live reads from the database. Each shows the real name and city of the signed-in user instead of hard-coded placeholder text. The tutor home additionally shows the verification status that the admin has assigned.

Figure 4.1 shows the screens of Admin, Tutor and Student homepages.

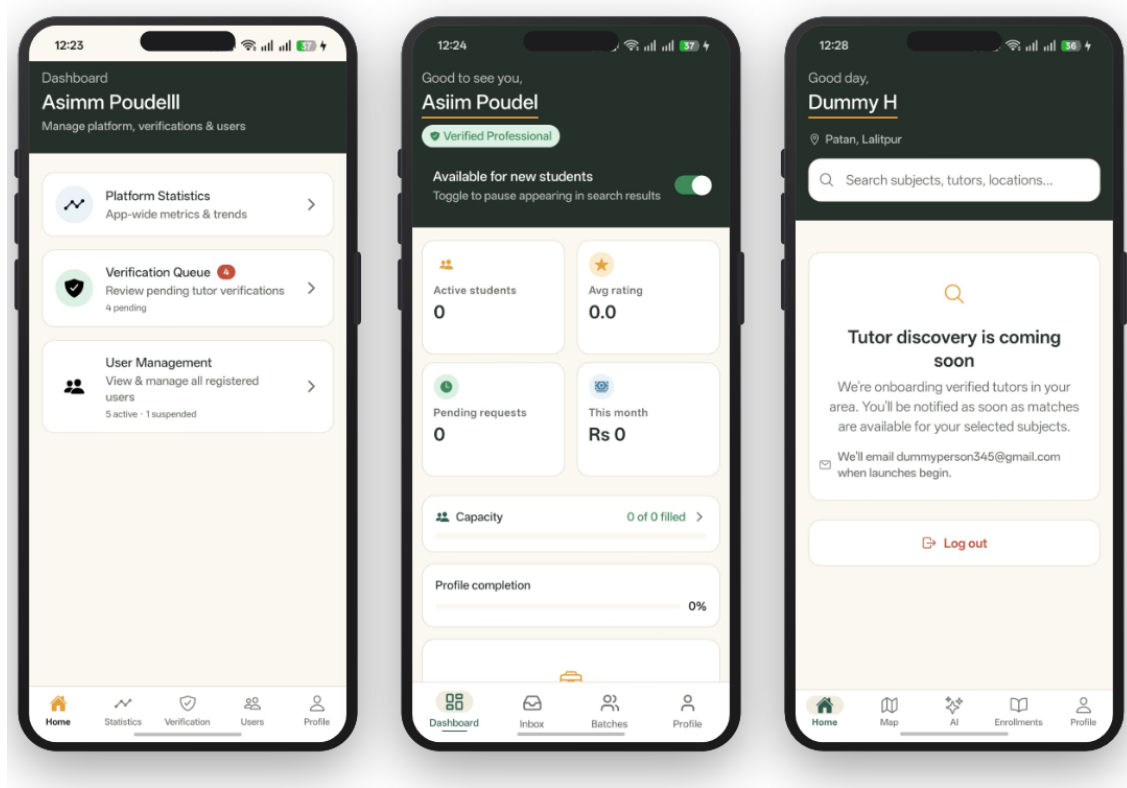


Figure 4.1: Dashboards

4.6 Profile Tabs

The student and tutor profile tabs share the same layout, which consists of an identity card, editable fields, menu rows, and a destructive log-out button. The profile tab allows the user to pick a new profile photo from the device library. A previously missing log-out affordance on the tutor profile tab has been added.

4.7 Admin Panel

The admin panel exposes a home view, a statistics screen, a verification queue, a user management screen, and an admin profile setup screen. Because the admin role is sensi-

tive, it is not assigned from inside the app. A small local admin script is used to grant the role to a fixed list of emails through the Firebase Admin SDK, which keeps the underlying admins collection locked to out-of-band writes only.

Figure 4.2 shows the user-management page on Admin Screen.

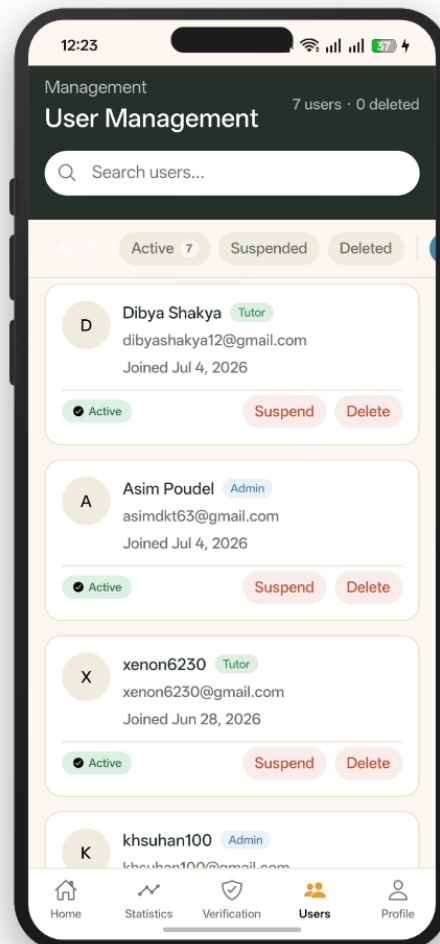


Figure 4.2: User Management Screen

4.8 Onboarding Visuals

The first two onboarding slides, which introduce the platform's discovery map and the AI chatbot, have been re-rendered against the new design palette so that they match the rest of the application.

4.9 Verification

The mid-term build passes a clean TypeScript type check and a clean ESLint run with zero errors. A manual end-to-end test on an Android device was carried out for the student, tutor, and admin roles.

5. TASKS REMAINING

The following are the tasks which need to be accomplished between the mid-term defense and the final defense.

- Tutor Search and Map Discovery
- AI Chatbot with Live Tutor Data
- Tutor Verification and Blue Tick Badge
- Enrollment Request and Approval Flow
- Ratings and Reviews
- Admin Dashboard with Real Platform Data
- Final End-to-End Testing and Demo Build

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